Keyword-Driven Testing

Overview

Keyword-Driven Testing (KDT) is a methodology in automated testing where test scripts are created using keywords representing actions or operations. These keywords are mapped to functions in the test framework, making it easier for non-programmers to contribute to test creation.

How It Works

1. **Define Keywords**:
    - Keywords are human-readable terms representing specific actions (e.g., "Login", "ClickButton", "VerifyText").
    - Each keyword corresponds to a function or method in the code.
2. **Prepare a Test Data File**:
    - Use a structured file format like Excel, CSV, or a database.
    - Each row typically represents a test step, with columns for:
        - **Keyword**: The action to perform (e.g., "EnterText").
        - **Locator**: The UI element to interact with (e.g., XPath or CSS selector).
        - **Input Data**: Values to input (e.g., username or password).
        - **Expected Result**: What to verify after the step.
3. **Develop a Test Framework**:
    - Implement a library of reusable functions (mapped to keywords) in the chosen programming language (e.g., Python, Java, or JavaScript).
    - A driver script interprets the test data and executes corresponding functions.
4. **Execute Tests**:
    - The framework reads the test data file, performs actions defined by the keywords, and logs results.

Advantages

- **Low Programming Requirement**: Testers focus on creating test data files, reducing the need for programming skills.
- **Reusability**: Functions tied to keywords can be reused across multiple tests.
- **Flexibility**: Supports modular and scalable test design.

Implementation in Python

Here's an example of a Keyword-Driven Testing framework in Python:

Keyword Mapping

```
def open_browser(url):
    driver.get(url)

def enter_text(locator, value):
    element = driver.find_element_by_xpath(locator)
    element.send_keys(value)

def click_button(locator):
    driver.find_element_by_xpath(locator).click()

def verify_text(locator, expected):
    actual = driver.find_element_by_xpath(locator).text
    assert actual == expected, f"Expected {expected}, got {actual}"

keyword_map = {
    "OpenBrowser": open_browser,
    "EnterText": enter_text,
    "ClickButton": click_button,
    "VerifyText": verify_text,
}
```

Test Data (e.g., Excel or CSV)

Keyword	Locator	Input Data	Expected Result
OpenBrowser		https://example.com	
EnterText	//input[@name='user']	test_user	
ClickButton	//button[@id='login']		
VerifyText	//div[@id='welcome']		Welcome, User!

Driver Script

```
import csv

# Example: Reading test data from CSV
with open("test_data.csv", "r") as file:
    reader = csv.DictReader(file)
    for row in reader:
        keyword = row["Keyword"]
        locator = row["Locator"]
        input_data = row["Input Data"]
        expected_result = row["Expected Result"]

        # Execute corresponding function
        if keyword in keyword_map:
            if expected_result:
                keyword_map[keyword](locator, input_data, expected_result)
            else:
                keyword_map[keyword](locator, input_data)
```

Python Example
Test Framework
python
Copy code

```
from selenium import webdriver

# Initialize WebDriver
```

```
driver = webdriver.Chrome()

# Keyword Function Definitions
def open_browser(url):
    driver.get(url)

def enter_text(locator, value):
    driver.find_element_by_xpath(locator).send_keys(value)

def click_button(locator):
    driver.find_element_by_xpath(locator).click()

def verify_text(locator, expected):
    actual = driver.find_element_by_xpath(locator).text
    assert actual == expected, f"Expected {expected}, but got {actual}"

# Keyword Map
keyword_map = {
    "OpenBrowser": open_browser,
    "EnterText": enter_text,
    "ClickButton": click_button,
    "VerifyText": verify_text,
}
```

Test Data (CSV Example)

```
csv
Copy code
Keyword,Locator,Input Data,Expected Result
OpenBrowser,,,https://example.com
EnterText,//input[@id='user'],test_user,
ClickButton,//button[@id='login'],,
VerifyText,//h1[@id='welcome'],,Welcome, User!
```

Driver Script

```
python
Copy code
import csv

# Read Test Data
with open('test_data.csv', 'r') as file:
    reader = csv.DictReader(file)
    for row in reader:
        keyword = row['Keyword']
        locator = row['Locator']
        input_data = row['Input Data']
        expected_result = row['Expected Result']

        # Execute the corresponding function
        if keyword in keyword_map:
            if expected_result:
                keyword_map[keyword](locator, expected_result)
            elif locator and input_data:
                keyword_map[keyword](locator, input_data)
            elif input_data:
                keyword_map[keyword](input_data)
            else:
                keyword_map[keyword]()
```

Implementation in Java

Using frameworks like Selenium and Apache POI for Excel data handling:

Keyword Mapping

```
public void openBrowser(String url) {
    driver.get(url);
}

public void enterText(String locator, String value) {
    driver.findElement(By.xpath(locator)).sendKeys(value);
}

public void clickButton(String locator) {
    driver.findElement(By.xpath(locator)).click();
}

public void verifyText(String locator, String expected) {
    String actual = driver.findElement(By.xpath(locator)).getText();
    Assert.assertEquals(actual, expected, "Verification Failed!");
}
```

Driver Code

```
// Example: Reading test data from Excel (using Apache POI)
FileInputStream fis = new FileInputStream("test_data.xlsx");
Workbook workbook = new XSSFWorkbook(fis);
Sheet sheet = workbook.getSheetAt(0);

for (Row row : sheet) {
    String keyword = row.getCell(0).getStringCellValue();
    String locator = row.getCell(1).getStringCellValue();
    String inputData = row.getCell(2).getStringCellValue();
    String expected = row.getCell(3).getStringCellValue();

    switch (keyword) {
        case "OpenBrowser":
            openBrowser(inputData);
            break;
        case "EnterText":
            enterText(locator, inputData);
            break;
        case "ClickButton":
            clickButton(locator);
            break;
        case "VerifyText":
            verifyText(locator, expected);
            break;
    }
}
```

Java Example
Test Framework
java
Copy code

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
```

```
public class KeywordFramework {
    static WebDriver driver;

    public static void openBrowser(String url) {
        driver = new ChromeDriver();
        driver.get(url);
    }

    public static void enterText(String locator, String value) {
        WebElement element = driver.findElement(By.xpath(locator));
        element.sendKeys(value);
    }

    public static void clickButton(String locator) {
        driver.findElement(By.xpath(locator)).click();
    }

    public static void verifyText(String locator, String expected) {
        String actual = driver.findElement(By.xpath(locator)).getText();
        if (!actual.equals(expected)) {
            throw new AssertionError("Expected " + expected + " but found "
+ actual);
        }
    }
}
```

Test Data (Excel Example)

Keyword	Locator	Input Data	Expected Result
OpenBrowser		https://example.com	
EnterText	//input[@id='user']	test_user	
ClickButton	//button[@id='login']		
VerifyText	//h1[@id='welcome']		Welcome, User!

Driver Script

java
Copy code

```
import org.apache.poi.ss.usermodel.*;
import java.io.FileInputStream;

public class TestExecutor {
    public static void main(String[] args) throws Exception {
        FileInputStream fis = new FileInputStream("test_data.xlsx");
        Workbook workbook = WorkbookFactory.create(fis);
        Sheet sheet = workbook.getSheetAt(0);

        for (Row row : sheet) {
            String keyword = row.getCell(0).getStringCellValue();
            String locator = row.getCell(1) != null ?
row.getCell(1).getStringCellValue() : "";
            String inputData = row.getCell(2) != null ?
row.getCell(2).getStringCellValue() : "";
            String expected = row.getCell(3) != null ?
row.getCell(3).getStringCellValue() : "";

            switch (keyword) {
                case "OpenBrowser":
                    KeywordFramework.openBrowser(inputData);
                    break;
                case "EnterText":
                    KeywordFramework.enterText(locator, inputData);
                    break;
```

```
                case "ClickButton":
                    KeywordFramework.clickButton(locator);
                    break;
                case "VerifyText":
                    KeywordFramework.verifyText(locator, expected);
                    break;
            }
        }
    }
}
```

Implementation in JavaScript (Cypress Example)

Cypress doesn't natively support KDT but can mimic it with modular functions:

Keyword Mapping

```
const keywords = {
    OpenBrowser: (url) => cy.visit(url),
    EnterText: (locator, value) => cy.get(locator).type(value),
    ClickButton: (locator) => cy.get(locator).click(),
    VerifyText: (locator, expected) => cy.get(locator).should('have.text',
expected),
};
```

Driver Code

```
Cypress.Commands.add("executeKeyword", (keyword, locator, inputData,
expected) => {
    if (keywords[keyword]) {
        if (expected) {
            keywords[keyword](locator, inputData, expected);
        } else {
            keywords[keyword](locator, inputData);
        }
    }
});

// Example: Define test case
cy.executeKeyword("OpenBrowser", null, "https://example.com");
cy.executeKeyword("EnterText", "input[name='user']", "test_user");
cy.executeKeyword("ClickButton", "button#login");
cy.executeKeyword("VerifyText", "div#welcome", "Welcome, User!");
```

JavaScript Example (Cypress)

Keyword Map

```
javascript
Copy code
const keywords = {
    OpenBrowser: (url) => cy.visit(url),
    EnterText: (locator, value) => cy.get(locator).type(value),
```

```
    ClickButton: (locator) => cy.get(locator).click(),
    VerifyText: (locator, expected) => cy.get(locator).should('have.text',
expected),
};
```

Test Case (JSON Example)

```
json
Copy code
[
    { "Keyword": "OpenBrowser", "Locator": "", "InputData":
"https://example.com" },
    { "Keyword": "EnterText", "Locator": "input#user", "InputData":
"test_user" },
    { "Keyword": "ClickButton", "Locator": "button#login", "InputData": ""
},
    { "Keyword": "VerifyText", "Locator": "h1#welcome", "InputData": "",
"ExpectedResult": "Welcome, User!" }
]
```

Driver Script

```
javascript
Copy code
Cypress.Commands.add("executeKeyword", (keyword, locator, inputData,
expected) => {
    if (keywords[keyword]) {
        if (expected) {
            keywords[keyword](locator, expected);
        } else if (inputData) {
            keywords[keyword](locator, inputData);
        } else {
            keywords[keyword](locator);
        }
    }
});

// Test Execution
cy.fixture("test_case.json").then((testSteps) => {
    testSteps.forEach((step) => {
        cy.executeKeyword(step.Keyword, step.Locator, step.InputData,
step.ExpectedResult);
    });
});
```